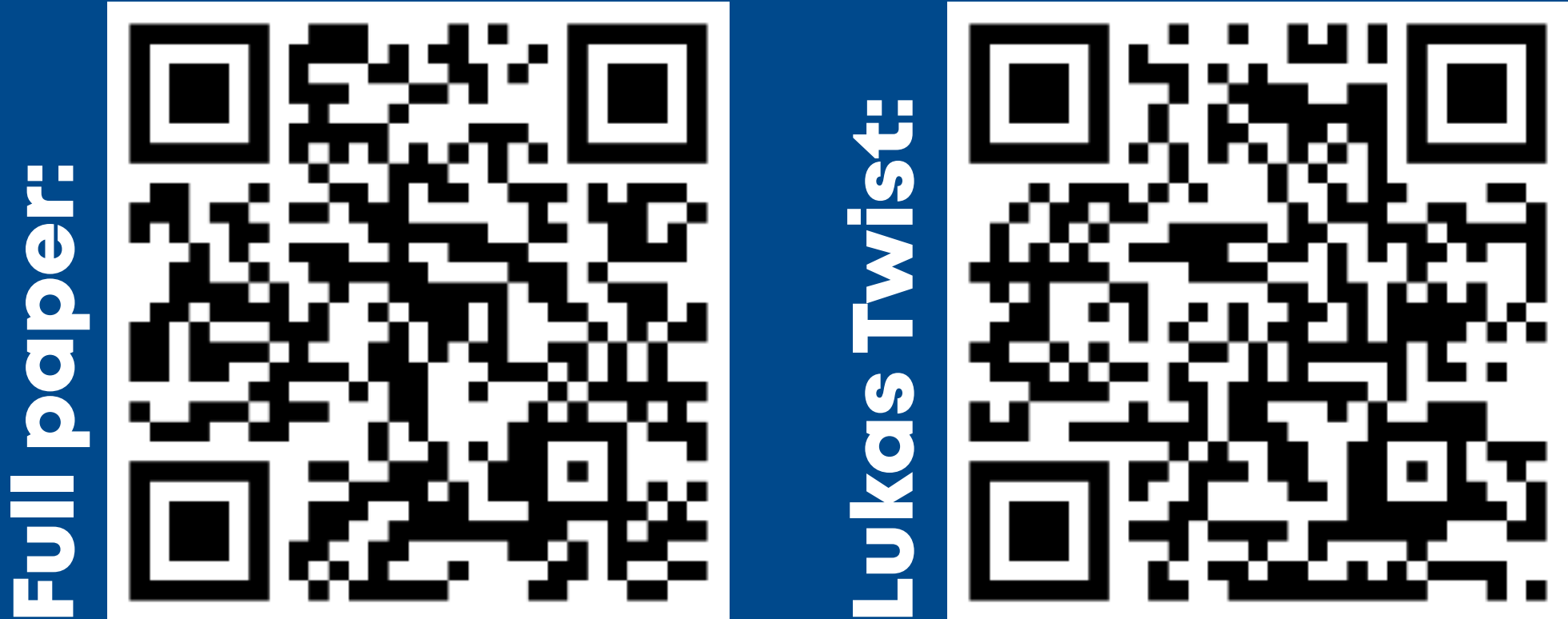


A Study of LLMs' Preferences for Libraries and Programming Languages

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LLMs Love Python: they overwhelmingly default to it, even for high-performance tasks where it is explicitly suboptimal. By favouring familiar defaults over actual task suitability, LLMs create negative consequences for security, performance, and open-source diversity.



Motivation

- Code-generation benchmarks mostly ask: does it work?
- But LLMs also make hidden design choices: which library? which language? safe, efficient or maintainable?
- When users omit technical details, *LLMs can silently steer technology adoption.*



Methodology

- Measured library and language choice in two settings: benchmark tasks and project initialisation.
- What do LLMs use when the technology is not specified?
- Do LLMs' natural language recommendations match what they actually use when generating code?

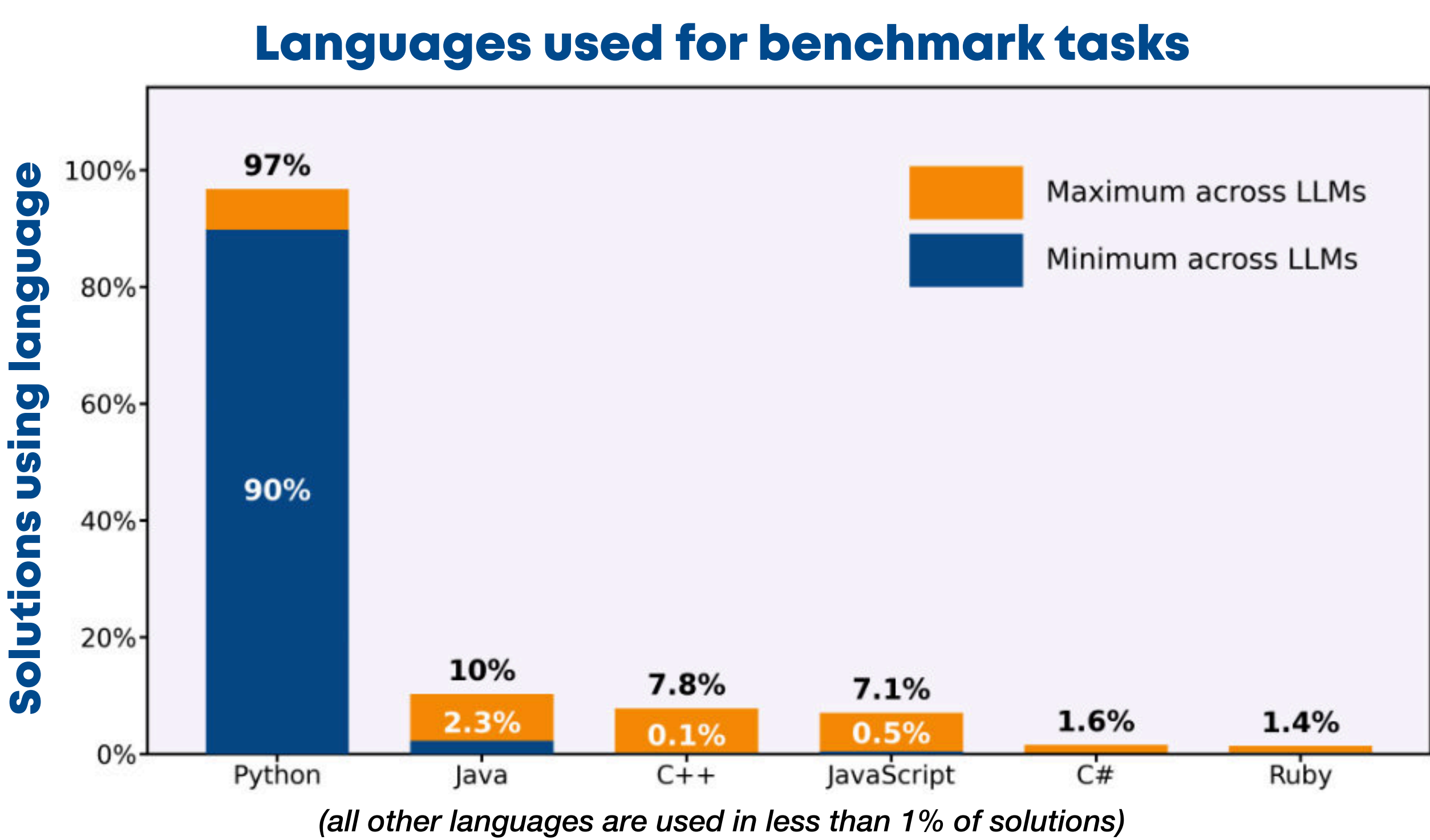
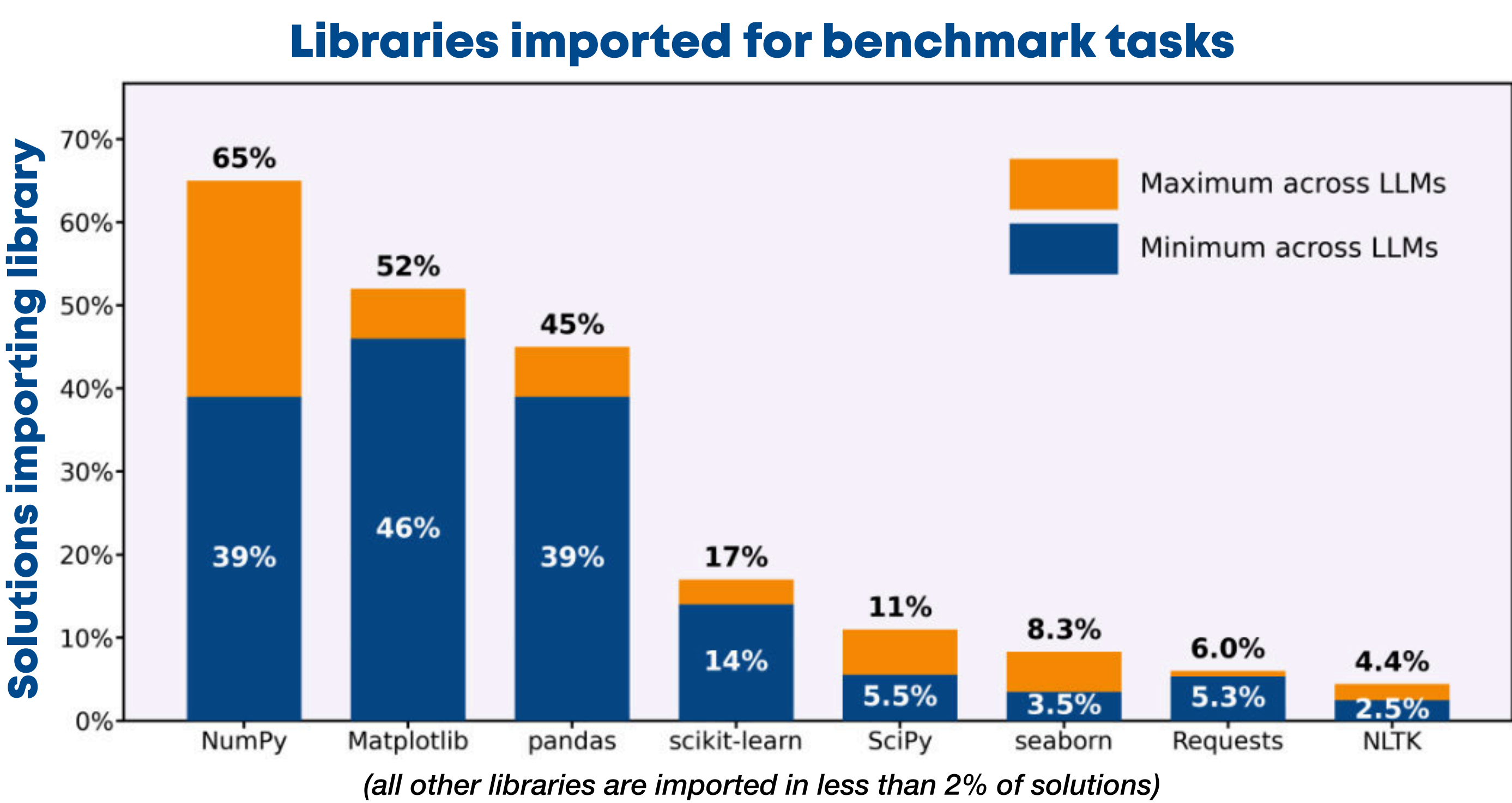


Library Findings

- The same few libraries dominate usage. Up to **45% of NumPy usage diverges** from ground-truth solutions.
- Each LLM used only 32–39 distinct libraries.
- Across tasks, *LLMs strongly favour older defaults over high-momentum alternatives*: pandas over Polars, Flask over FastAPI, Dask over Ray/Celery.

Language Findings

- *Python dominates* generated solutions across all models, accounting for 90–97% of benchmark task solutions.
- Even in performance-critical project prompts, Python remains the most-used language in 58% of cases.
- LLMs *consistently contradict their own recommendations* for new projects, rarely choosing C, C++, Go, or Rust.



1. Security threat.

Suboptimal defaults can steer production systems toward unsafe or inefficient implementation choices.



2. Ecosystem impact.

LLM preferences may reinforce dominant libraries, reducing visibility for emerging open-source tools.



3. Evaluation gap.

Code evaluation should measure whether technology choices fit the task, not just whether code runs.

